

The selected δ -floor

one lemma: $\liminf \delta_k > 0$;
preferred on a slice, not proved

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Abstract

r442 wrote $\kappa^\dagger = 1\{q^\dagger > 1\}$ with $\delta = 1 - q^\dagger = -\text{sch}$. The observed rise of $q^\dagger$ toward 1 on Selected is exactly the old r421 question “does $R = -\varphi_{\text{bb}}$ go to 0 or to a floor?” in the last-pivot coordinate. The one lemma of this round is $\liminf \delta_k > 0$ on the selected sequence. It is *not* proved. The r417/r418 charts are SATZ; a floor is preferred only on the r421 $k = 5..9$ slice; the full selected sequence does not independently prefer it; k_z16 sits far below. No RH claim.

Status. The chart identities of Lemma 2 are **SATZ**. Propositions 1–2 are a named AIC census and named kills. The $\liminf$ itself is *open*. Ausgang REDUZIERT / CHART_EXACT / SLICE_FLOOR_PREFERRED / FULL_SEQUENCE_UNSETTLED / KZ16_SELECTED_SPECIFIC / DEEP_LOCKED / COFINAL_OPEN. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The lemma, and the chart

Lemma 1 (Selected δ -floor). *On the selected sequence $a_k = 2^k$, $\liminf_k \delta_k > 0$.*

Not proved this round. The working identity is:

Lemma 2 (Chart). $\delta = -\text{sch}$. *On the vacuous branch $\delta = R + \tau^2$ with $R = -\varphi_{\text{bb}}$. On the P1 branch $\delta = R + (b_{\text{un}}^2 - a_{\text{un}}^2)$. In both cases $R = 2 - \text{den} - \Sigma$.*

Proof. r433 and r417, over $\mathbb{Q}$: P1 sch = $-2/3$, VAC sch = $-7/6$. On the six live selected windows the chart residual is 0. $\square$

Selected $k = 6, 7$ are vacuous; the rest are P1. τ^2 is already $O(10^{-3})$ on VAC and falls $0.019 \rightarrow 0.0006$ on P1, so at $k = 9$ one has $|\delta - R| = 3.5 \cdot 10^{-4}$. den is flat in $[1.48, 1.70]$ ($O(0.5)$ room). $\Sigma = 2 - \text{den} - R$ wanders $0.211 \rightarrow 0.480$. The decay of δ is the decay of R .

2 Fits, both abscissae

Three diagnostic models (r421, not a proof): M_1 floor + ck^{-a} , M_2 pure power, M_3 $c/\log$.

Proposition 1 (Slice vs full sequence). *On $k = 5, 6, 7, 9$ (the r421 slice, $k = 8$ not rebuilt), M_1 wins for δ with $\delta_\infty = +0.0267$ (ΔAIC vs M_2 is 6.0). The same slice on R gives $R_\infty = +0.0291$ ($+0.0298$ with the r421 $k = 8$ pin). On the full selected $k = 3..9$, M_3 wins and M_1 hits the grid floor -0.02 (the $k = 5$ bump 0.109 wrecks a global floor). Against N , M_2 wins: N is not monotone ($k = 5$ has $N = 96 < k = 4$'s 184) and is an illegal abscissa (r427). Drop- $k = 5$ flips to M_2 ; drop- $k = 9$ keeps M_1 but jumps δ_∞ to 0.045.*

EXT-6 is the second line: $\delta \in [0.0264, 0.0452]$, mean 0.0357, sitting at the slice floor. $k = 10$ (k_z197 , $N = 4071$) times out at 20s inside `border_chain_pack`: there is no C yet, so a CG solve of $(I - C)b$ cannot unlock the point. $k = 11, 12$ skipped.

3 Conditional close, and its honesty

If a floor $\delta_\infty > 0$ exists then, eventually on Selected, $q^\dagger \leq 1 - \delta_\infty \approx 0.973$, hence $\kappa^\dagger = 0$ by r442, hence the frequently-mincut is filled. That implication is tautological. The missing input is existence of the liminf. It is now a scalar sequence, not a matrix cone — but it is still a census ($k = 5.9$ plus EXT), not a theorem. A provable lower bound would need $\Sigma_\infty < |c_{J,\infty}|$ with $c_J = \text{den} - 2$. The den-room is $O(0.5)$ and near a theorem via $\gamma < 1$ (r424: $\|b\|^2 \leq S$ is SATZ; $S < B_w \Leftrightarrow q_N < 1$). q_N is *not* $q^\dagger$ (r433); that bridge is the circularity gate.

4 kz16 and kills

Proposition 2 (k_z16). *The tightest living old-ladder window has $\delta = 0.00151 \ll 0.027$, with $R = -0.003$ already negative (life is a τ^2 gift). k_z12 is the same pattern ($\delta = 0.00594$, $R = -0.016$).*

A floor that covered the old ladder would be false. The slice floor is Selected-specific, as r428's quantifier artefact already suggested. Dead $\chi_3\text{-}15$ has $\delta = -0.023 < 0$. Soft mutants (tr G , moments) remain blind (r440/r442).

Claim boundary

Nothing here is evidence for or against the Riemann hypothesis. Lemma 2 is finite algebra. Lemma 1 is open.

References

- [1] Round 417, Sylvester chart.
- [2] Round 418, $\varphi_{\text{bb}} = c_J + \Sigma$.
- [3] Round 421, R -floor AIC.
- [4] Campaign audit: N is not monotone.
- [5] Ladder criticality as a quantifier artefact.
- [6] $\delta = 1 - q^\dagger = -\text{sch}$.
- [7] $\kappa^\dagger = 1\{q^\dagger > 1\}$.